

Digital Sovereignty: What it is and why it matters for HCI

Dennis Lawo

Information Systems, University of Siegen
Germany
dennis.lawo@verbraucherinformatik.de

Margarita Esau-Held

Information Systems, University of Siegen
Institut für Verbraucherinformatik, Hochschule
Bonn-Rhein-Sieg University of Applied Sciences
Germany
margarita.esau@uni-siegen.de

Thomas Neifer

Information Systems, University of Siegen
Institut für Verbraucherinformatik, Hochschule
Bonn-Rhein-Sieg University of Applied Sciences
Germany
thomas.neifer@h-brs.de

Gunnar Stevens

Information Systems, University of Siegen
Germany
gunnar.stevens@uni-siegen.de

ABSTRACT

In recent years, digital sovereignty has become a central term in digital policy discourses. Both authoritarian and democratic states use digital sovereignty as a base for their digital policy, although, the individual interpretation and resulting policy and power balance might be quite different. Given the importance of this term, the HCI community and the users as the core of our research would benefit from taking up the discussion by finding own definitions of human-centred digital sovereignty, contributing to policy discourses, and to strengthening the position of users under non-sovereign conditions. This paper aims to initiate and provoke such discourse within the community by (1) introducing the policy term to HCI and providing an overview of how it is used, (2) arguing for the relevancy of the term, and (3) proposing possible ways forward.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI);

KEYWORDS

Digital Sovereignty, Individual Empowerment, Consumer Protection, Privacy, Policy

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1 INTRODUCTION

"Digital sovereignty, and the fight for it, touch everyone, even those who do not have a mobile phone or have never used an online service." [30]

In recent years, *digital sovereignty* has become a central term in digital policy discourses [44, 64]. The term is used as a leitmotif for a digital agenda in both authoritarian and democratic states. Depending on the different political orientations of the nations and supranational organisations, the term is interpreted differently and a different policy is derived from it. However, the central idea is that states want to assert their sovereignty - traditionally their control over territory and people - in the digital space vis-à-vis other states, corporates, and individuals [44, 64].

The evolution of the Internet and the accelerating digital transformation are challenging nations, which usually exercise sovereignty over a clearly defined territory, to exercise that same sovereignty on the untamed Internet. As a countermeasure to a possible loss of control in the digital sphere, states have established various technical and legal measures to restore sovereignty [30]. At the same time, the public has been convinced of the importance of (supra)national agendas, that control of the digital is good and useful in terms of governments [25, 64]. It is to be expected that the discussion about sovereignty will intensify with the development and spread of Artificial Intelligence (AI), Internet of Things (IoT), as well as increasing connectivity [30, 64].

Given the importance of this term within the political sphere, it is surprising that the HCI community did not take up the discussion. We can expect that the policy and regulations that emerge from the discourse will have a strong impact on individuals [30, 44, 64]. For example, for users, when their privacy is protected or disregarded, or their rights are restricted or protected from technological surveillance. In addition, it also affects non-users as already today digitisation has an impact on their privacy, their access to products and services, their practices, and the societies they live in [30].

The policy discourse is mainly driven by strong actors, states, and companies [44]. Even though some place citizens and users at the core of their efforts [44, 64], the outcome might not be positive [30], especially if the focus is on protecting exclusive societies and not all people, e.g., when colonialist patterns are reembarbed [4].

Against this background, as a community focusing on humans, users, citizens, and consumers, we should take up and critically accompany the policy discourse and actions taken. The HCI community and the users as the core of our research would benefit from taking up the discussion by finding their own definition of human-centred digital sovereignty, contributing to and participating in policy discourses, and strengthening the position of users under non-sovereign conditions, e.g., through interventions.

This paper addresses the little HCI research on and involvement in the digital sovereignty discourse along three simple guiding questions:

- (1) What is digital sovereignty?
- (2) Why is digital sovereignty important to the HCI community?
- (3) What should we do in the face of political ambitions to strengthen a human-centred agenda?

To answer these questions, this paper introduces the policy discourse and uses it to specifically address the demands for a human-centred agenda and individual empowerment. In addition, this paper reviews recent work in the form of a literature review [68], which can be found in the ACM Digital Library under the term *digital sovereignty*. It analyses the use of the term in these papers, as well as the cases and perspectives, to provide a starting point for a broader discourse within the community. Finally, we will argue for the relevancy of the term, and propose possible ways for action.

Inspired by the legacy of essays that initiated and provoked new lines of research (see, e.g., [5, 6, 13, 66]), this paper aims to initiate and provoke a discourse within the HCI community. The contribution of this paper is not a stocktaking of research from privacy to digital literacy — fields being relevant for the discourse — but an introduction to the policy and HCI discourse on digital sovereignty, arguments for why it matters, and potential ways forward. As such, this paper serves as a starting point or prequel for (hopefully) a variety of research contributions.

2 WHAT IT IS

2.1 The Policy Discourse

"Today, sovereignty always primarily means a state's independence vis-à-vis other states (external sovereignty) as well as its supreme power to command all powers within the territory of the state (internal sovereignty)."[64]

Before considering the prefix *digital*, we will introduce sovereignty as a general idea. The above definition of sovereignty is based on Jean Bodin's definition from the 16th century, which describes the ability of the ruler, the sovereign, to make a decision and execute it [64]. The ruler or sovereign is, of course, depending on the context, to be replaced by various bodies, e.g. parties, dictators, or the people. Within such a definition of sovereignty, the territory of the state or over which the sovereign rules plays an important role [32].

In the digital sphere, there is no such clearly defined territory, even if some states are trying to establish (partially) closed networks. Thus, this characteristic of the digital challenges the idea of state sovereignty from various ideological standpoints. For example, from a cyber exceptionalism perspective, it is argued that the digital sphere is so different from previous objects of state control that this sphere should also be treated differently. Often the argument is, that

states as agents in the digital sphere are too slow and inefficient in managing the fast pace [64]. Often, this perspective is linked to libertarian attitudes that reject sovereignty as a political interference in favour of the free development of individuals and businesses [41]. Such arguments can be seen in debates about crypto currencies, as means of payment not controlled by the state [60].

Another example is the multi-stakeholder Internet governance perspective. The leading idea is that the stakeholders should manage the Internet on their own. The state would be a partly non-sovereign actor and would deliberately forego this sovereignty to assign the task of norm-setting, regulation, and process definition to non-governmental bodies, often from the technology community. The result would be decentralised management and negotiation of best practices [22, 43] between states and NGOs. However, it must be acknowledged at this point that internal disputes and a lack of coordination have so far not been able to achieve a sustainable success of decentralised approaches in contrast to national regulation. Also, many states do not allow their sovereignty to be (completely) questioned by other parties and try to exercise state sovereignty through multilateral and bilateral agreements as well as international organisations [64].

The claim to be the sovereign is not limited to states, but also individuals and corporates claim to have and to be able to exercise certain sovereignty. Resulting of this contested sovereignty over the digital, different constellations of actors, with different goals and capabilities, in different political contexts, we can argue that the digital sphere, and especially the Internet is heading towards a phase of claiming and negotiating sovereignty with an unclear and potentially threatening outcome. Some authors even name this situation "tech war" [23]. However, we can assume that state actors will be exercising a certain sovereignty over the digital sphere, or at least try to exercise it in the future. Some states' plans rely on the idea of a strong and sovereign state, which can exercise sole control over its part of the Internet, while other states promote digital or surveillance capitalism in a laissez-faire approach to benefit from economic growth [39]. Again other states are heading for a third way, with a strong emphasis on the individuals as users, consumers, and citizens [30, 44, 64]. Others lack the power to exercise sovereignty at all [4].

In the following, these different actors and the discourse associated with them will be explained [44, 64].

2.1.1 (Supra)National Sovereignty. Given the historical connotation of sovereignty and the nation as the traditional sovereign of the last century, it is not surprising, that the most prominent actor to claim digital sovereignty is the nation.

As already indicated in different political contexts, sovereignty comes with different political claims. Pohle et al. [64] identified two different lines of thinking. On the one side, authoritarian and non-liberal governments see digital communication as a threat to their political systems [64]. For example, China developed a concept of sovereignty to counteract the threat of a loss of media control with the uprising of the Internet and social media [26]. Another example is Russia, whose government attempts to gain control over a national part of the Internet, while still benefiting from other parts of global connectivity [70].

But also liberal democracies thrive for an agenda of digital sovereignty. In those contexts, the ideas of sovereignty, however, are mostly driven by security concerns, as global connectivity revealed vulnerabilities and threats to infrastructures relying on the digital sphere [64]. At this point, we want to highlight that often the authoritarian or semi-authoritarian states are seen as a threat based on their policy, but as we can see in the work of Möllers et al. [53] that also the liberal-democracies take actions which are in conflict with their ideal and might endanger civic societies and understanding of democracy.

Based on both motivations, the actions are taken in the direction of binding data and processes to a local network under the jurisdiction and protection of the nation or supranational organisations, e.g., the EU [44], with limited access of other actors, e.g. other nations or companies [64]. Thereby, the protection of data and the security of infrastructures is not the only motivation, but the increased access to data of citizens, e.g., for surveillance reasons, and potential economic benefits from closed innovation systems [21, 35, 54].

2.1.2 Economic Sovereignty. Another part of digital sovereignty, which is also used by both democratic and authoritarian states, motivates national independence from other actors from an economic perspective. These actors can be governmental, but also economic, e.g., companies especially monopolist service providers. The national independence already shows that there is an overlap with the previous section, which can also be observed in the particular measures taken by the states [7].

It should be noted, however, that the narratives and motivations differ, where it is often about reducing dependence on other countries and their companies, e.g., China and the U.S. as dominant market players [72]. Based on this motivation, economic digital sovereignty can also be seen as part of larger efforts to digitise and regulate trade flows, especially through digital channels [64], and also to strengthen local economic actors [17].

Another line of argumentation is based on the creation of local services, e.g., own cloud providers within the framework of the European Gaia-X project [3, 16], to enforce the own standards of consumer and data protection [35, 52]. In the context of emerging economies, especially countries that have suffered from colonisation, economic independence, and similar attempts are also discussed against the background of digital colonialism. The goal is to reduce dependence on Western countries and establish their infrastructures [4].

2.1.3 Individual Sovereignty. This part of digital sovereignty is certainly most relevant to the HCI community, due to its strong focus on individuals (and their civil societies) in their various life contexts.

Sovereignty, here, focuses on individual self-determination in the digital sphere and the autonomy of individuals (and civil societies) as citizens, employees, users, and consumers [44]. This part of the political discourse takes place largely in democratic contexts, with a certain prevalence in Europe [44, 62–64]. While again there is strong overlap with the other perspectives, see for example the consumer protection argument in the context of economic sovereignty, the discourse is partly characterised by a counter-model that places

sovereignty in the hands of citizens rather than in governmental institutions [44, 64].

While the above introduction to the idea of digital sovereignty still centres the state as an actor with a positive contribution to digital sovereignty, there are of course also critical observers and activists within social (technical) movements who see the state (and here not only, but also the democratic states) as a threat to the digital sovereignty of the individual users, consumers, or citizens [25].

From a perspective of the measures and actions taken, however, the two positions do not differ greatly. While the former perspectives see the state as having a greater responsibility to protect its citizens as consumers of services, the latter movement places citizens at the centre and assigns responsibility for the appropriation of suitable technologies and the protection of their data [25]. There is a strong focus on measures to increase digital literacy, data protection, and security, but also user-friendliness and transparency of algorithms and business models [44, 64].

2.2 The HCI Discourse

In the following, we will give an outline of how the digital sovereignty discourse has been adopted by the HCI community. For this purpose, we conducted a literature survey based on research from within the ACM Digital Library.

In the first step, we searched for the term *digital sovereignty* without any further restrictions (Query Syntax: "query": { AllField: ("digital sovereignty") } "filter": { ACM Content: DL }; Query Date: January 13 2023). This results in 40 papers, primarily coming from other communities and focusing on non-human-centred issues. However, the search clarifies two things. First, we see how the term appeared in two publications in 2015 and has been steadily growing in importance since then, with 22 publications published in 2022. Second, we see how a large number of publications deal with the dominant topics of digital sovereignty. Examples include European cross-border data sharing [67], Industrial Data Space [1], AI policy [61, 79], blockchain [18, 42], tribalism, shielded, or monopolised Internet [28, 29, 74], and e-Government [9, 47, 55, 58]. Moreover, the geographic context of the studies shows that the discourse goes beyond Western countries, e.g., Brazil [9], the Kyrgyz Republic [11], Russia [58], or Central Asia [78].

Based on a refinement of the survey, that filters for higher level CSS concepts related to our community (Query Syntax: "query": { AllField: ("digital sovereignty") } "filter": { ACM Content: DL, CCS 2012: Human computer interaction (HCI) OR Human and societal aspects of security and privacy OR Human-centered computing OR Computer supported cooperative work }; Query Date: January 13 2023) we see that 10 publications locate themselves within the broader HCI community. Within these publications, the research is much more focused on the themes around individual empowerment, ranging from designing transparent e-Government websites [77] or smart cities [8], service switching intentions within the GDPR legal context [49], the privacy of used smart devices [12] and services [2], technology independence from monopolists [31, 33, 34, 51], to the effect and design of social media in the context minimising user autonomy and data exploitation [65]. The research thereby is both empirical as well as design-driven. Although in the context

of GDPR [49], the policy is used as motivation for analyses and design, the paper does not address policy implications directly. This is different from the first query, where more research focused on policy implications, for example, revealing best practices for blockchain regulation [42].

Within this research, no holistic definition of digital sovereignty is given. Therefore, in the following, we want to locate the current narratives and attempt further defining the term for HCI.

The broadest definition is given by Luzsa et al. [49] who understand "the term 'digital sovereignty' [as the] answers to these threats and ways to increase privacy, security, and informational self-determination of individuals, economic actors, and democratic institutions". Whereby the threats for individual users, e.g., manipulation or identity theft, companies, e.g., intellectual property theft, and societies as a whole, e.g., polarisation and mistrust, are caused by fake news, hacking, and monopolisation of services.

While this definition accounts for the whole network of actors within the political discourse, other authors, such as Wessel et al. [77] see individuals being digital sovereign when they can "act deliberate and with self-determination". Essential for this is the ability to understand which personal data is shared with which governmental or commercial actor. Quite similarly Berger Bernardes et al. [8] locate digital sovereignty in the discussion of privacy and control over their data. Their case is situated around smart cities in Europe that apply GDPR. The commercial line of argumentation for digital sovereignty is used by Birnstill et al. [12] who argue that sovereignty users are sovereign in the context of smart assistance when they "able to determine which data is collected and used for which purposes".

Rafael Grohmann [33] also argues for digital sovereignty in the context of data control and ownership. But his arguments go beyond demanding transparent digital infrastructures, but for autonomously designing and maintaining own infrastructures to be independent from monopolists and their infrastructure and algorithms, which might enforce inequalities and certain ways of working. Given the context of Brasil, narratives of digital colonialism and dependence on Western countries are also picked up. In the context of smart cities, Heitlinger et al. [34] similarly argue for software and hardware autonomy and the inclusion of citizens in the ownership, design, and maintenance of the systems to counter "the abstract digital spaces and proprietary infrastructures of neoliberal smart cities". They also address the need for skills and literacy to enable such self-governance of a digital-common. Similarly, Marichal et al. [51] locate their work on open-source mixed-reality environments within a broader discourse of digital sovereignty. However, the dependence on monopolists is not elaborated on. Other research [31] cites digital sovereignty as one of the benefits of their recycling platform, which decreases the dependence on (foreign) resources and producers.

Schelenz et al. [65] understand digital sovereignty in the context of protecting users against the power of social media platforms and regulating these. They argue, that the service providers "minimiz[e] users' autonomy, marginaliz[e] minorities, and exploit[e] users' data for profit maximization". This critique on social media platforms also relates to the first definition, which highlighted the threat coming from fake news [49].

Although the corpus of literature picking up the term *digital sovereignty* is limited, we aim to provide a (preliminary) definition of human-centred digital sovereignty that reflects the above narratives — (1) Sovereignty over (Personal) Data, (2) Sovereignty over Algorithms and Software, (3) Sovereignty over Hardware and Infrastructures, and (4) Sovereignty over and based on Literacy and Information.

Human-centred digital sovereignty emphasises the right, literacy and resources of humans — as citizens, consumers, or users — to autonomously access and control data, software, hardware, and information in alignment with their values and interests. Being sovereign requires *access*, independent from other actors' interests, to technology and (factual) information as the basis of participation in modern society. Moreover, it entails *control* over data collection, storage, and sharing, and the adoption, usage and outcome of technology. *Literacy* is fundamental to make these informed decisions about data and identity, assessment and use of algorithms, and the selection, maintenance or repair of hardware and infrastructures.

3 WHY IT MATTERS

Based on this understanding, the following will focus on the benefits of engaging in the discourse, how it is linked to our community, and potential ways forward.

3.1 Interrelated Policies and Threats

First of all, we want to emphasise the ubiquitous dimensions of digital sovereignty. Individuals are affected by the loss of control lack of access, and limited literacy in all aspects of life. Increasing surveillance [45] and tribalisation of the Internet [29] endangers the positive visions of this community [29, 39]. Technology becomes a surveillance nightmare, makes dependent instead of providing support or remains an unattainable dream due to a lack of resources [4].

A very fitting example is given by Luciano Floridi [30]: European countries, including Germany [48], Great Britain [19], and Italy [27], were forced to scrap their ideas for a Corona app to be compatible with the decentralised systems of two companies, Google and Apple. Moreover, according to Floridi [30], Apple announced to have the possibility to shut down the solutions when no longer needed. What we can see in this example is that even governmental actors have a decision forced on them by companies, both technically and in terms of health policies to contain the pandemic. Furthermore, it should be possible for everyone to imagine the potential impact on the health of individual citizens as no access to alternatives or control over the app itself was given.

Due to this ubiquitous dimension and the dependencies of entire states, companies, and individuals, the discourse is certainly different from other trend topics. Even if these topics come with disrupting impact, the "tech war"[23] on digital sovereignty poses fundamental questions about the continued existence and future of the digital sphere as we know and envision it [39]. Thus, to understand and empower individuals' digital sovereignty we need to combine different perspectives holistically and think of sovereignty as a value within our research and design. For example, for the above case designing and researching a sovereign app would require negotiating privacy, e-health, UX, and e-government perspectives. In

this sense designing and design research needs to prioritise digital sovereignty as an own value [69]. The goal of these considerations should be to "pursue distributions of digital sovereignty that provide the best balance for each [moral] calculus" [24], for example for the above case, the value of privacy vs. democratic values.

Thereby, it is important to not just continue the same research, but collect and compare cases of sovereignty and non-sovereignty to understand the topic holistically and develop a clear concept of what human-centred digital sovereignty is. Such a concept would enable us to, first, locate the sovereign in a certain digital context [54] and second reflect on the different perspectives and influences on control, access, and literacy to make better decisions about their distribution [24]. Locating the sovereign, within its digital – but also social and cultural – context thereby requires consideration of how the value of sovereignty plays out in different contexts [14], without falling for the trap of cultural relativism as an argument for the non-sovereignty of individuals.

Emphasising digital sovereignty when designing ultimately demands us to understand individuals' needs and capabilities for digital sovereignty as we need to strive for realistic distributions within our design. Thereby, this demand also affects the design context, as we will (probably) always be dependent on certain companies or policies. We should critically research the impact on sovereignty coming from different forms of collaboration and develop new forms.

3.2 Sharing Best-Practices, Knowledge, and Software

HCI attempts should of course focus on empowering the individual against emerging dystopia such as surveillance capitalism (monopolistic/sovereign corporations) [20, 45] or social scoring/rating (sovereign surveillance states) [50]. For example, open-source/free software technologies to strengthen states and their societies to reduce dependence on monopolist technology providers [4, 56, 71], to support individuals against their government, e.g. to bypass censorship on the Internet [15], or to share and compare policy intentions under a common lens [76].

With a human-centred understanding, we need to foster user participation and engagement to place this positive vision and empower individuals, whether in collaboration or shared sovereignty with other actors (e.g., policies focusing on consumer/user protection [44]), a strong focus on software communities and free software [71], or in cases of oppression based on autonomous digital infrastructures [15]. Thereby, participation is particularly important, as it is not about our ideas of sovereignty, but the sovereignty of the people affected. Any other understanding, would not empower the individuals, but raise new dependencies, although the actions were well-intended.

It is certainly important to take stock of the research within the community (see, e.g., [6]). We already have a great variety of knowledge, participatory design approaches, and best practices on many topics ranging from open-source/free-software, usable-privacy, to grass-root movements. As a first step, it might be worth discussing those under the new outlook of ongoing fights for sovereignty and adapting them to decrease dependencies with other actors and reflect on our relative sovereignty vis-à-vis our users and participants.

3.3 Visibility and Informing Policy-Makers

To advocate user rights and interests, engaging with users in design might not be enough given the power distributions. We also need to foster visibility of empirical and design contributions in the policy discourse and the active transfer into this discourse. In the past, the HCI community has already proven that it can influence policy discourses [36]. Many examples can be found in the context of dialogue design and accessibility standards [36, 46]. For such policy contributions, however, it is necessary to actively engage with the policy discourse [37], such as critically accompanying the implementation of GDPR [59] or uncovering dependencies, e.g., Nouwens et al.'s work [57] on Danish knowledge workers' dependency on technology monopolists.

For success, however, we need to emphasise communication more strongly [46]. From this perspective alone, we should adopt the concept of digital sovereignty, as it addresses current policy discourses and thus enables policymakers to discover and relate to our contributions. Von Grafenstein [75] and Jakobi et al. [38] resemble another factor, which is the interdisciplinary exchange, e.g., with the legal and policy community. This enables a direct exchange with scholars at the front of recent developments. This might support finding a common understanding and ultimately provides a new channel to disseminate HCI contributions.

3.4 Research on, around, under, and with Digital Sovereignty

Finally, we should anticipate the influence and consequences of certain policies on our work. Already today, the political context strongly influences the work of HCI researchers and designers [40], we are dependent on funding, but also on a legal framework that allows us to conduct empirical research and deploy prototypes [46]. For example, with a lack of access to systems and networks, the work of colleagues worldwide will be hampered.

In addition, also economic actors are creating dependencies already today and they could increase in the future. For example, the dependencies on social media platform operators, that have control over the access to data, algorithms, and interfaces [10, 73] or on cloud providers, when designing interventions. In addition, just like the Danish knowledge-workers [57], we are dependent on a few companies that develop the software we work with, e.g. Microsoft Word, as probably the primary software to write papers like this.

Thus, we need to anticipate and critically accompany the impact on our work. This should entail working on our sovereignty, e.g. by developing sovereign research tools and artefacts or informing policy as well. Important, however, is the consideration of our impact on the sovereignty of individuals, to not act in our interest only.

4 CONCLUSION

While the ideas of engaging in policy discourses are not new to the HCI community and in particular for the digital sovereignty discourse a lot of subfields within HCI are already addressing questions at the forefront of policy and design, this paper argues for picking up the policy discourse more directly. In particular, this paper highlights what is discussed in the policy sphere, how HCI already started to pick up the topic, and why we should engage

even more. We argue that digital sovereignty is not just another trend topic, but a topic that is so interrelated with the networked world and that certain actors are questioning what this community is envisioning for the future of the digital sphere. Future research, should pick up the discourse, elaborate on human-centred digital sovereignty, engage with policy-makers and researchers, design open-source/free-software tools to support those living under non-sovereign conditions, and ultimately envision alternative futures with empowered users and a reduced detrimental and involuntary dependencies.

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